

1. Conceptual Data Model

- Entities:
- Customer
- Represents individual or business customers providing orders.
- Order
- Represents purchase transactions tied to customers.
- OrderSummary
- Joined information of customers and their orders, including SCD Type 2 history.
- CustomerAggregateSpend
- Aggregated daily spend per customer by name.
- Relationships:
- One Customer has many Orders.
- Orders are summarized and joined into OrderSummary with historical tracking.
- OrderSummary aggregates into CustomerAggregateSpend per Name and Date.
- Business Definitions:
- Customer: An identifiable entity involved in orders.
- Order: A transaction or purchase with product and quantity details.
- OrderSummary: A historical, joined dataset from customer and order with SCD Type 2 logic.
- CustomerAggregateSpend: Sum of daily spending per customer name.

2. Logical Data Model

- Customer
- CustId (Primary Key)
- Name
- EmailId
- Region
- Order
- OrderId (Primary Key)
- ItemName
- PricePerUnit
- Qty
- Date
- CustId (Foreign Key → Customer.CustId)
- TotalAmount (Computed, $\text{PricePerUnit} \times \text{Qty}$)

- OrderSummary
- CustId
- Name
- EmailId
- Region
- OrderId
- ItemName
- PricePerUnit
- Qty
- Date

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Technology-Agnostic Notes

- No physical storage platform is assumed beyond use of Delta-compliant tables
- Table and field types correspond to generic RDBMS and Delta conventions
- SCD Type 2 logic uses generic fields: StartDate, EndDate, Active—field definitions are configurable per implementation style

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